

Electrical Instrumentation Part-II

PowerPack Phase 2

Phase Locked Loop (PLL) Mastery

Reverse-engineered semester preparation notes
Diagram-safe, derivation-first, exam-answer style

Semester Command

This Phase 2 PowerPack is for mastering every PLL-type semester question. Learn the answer in this order:

Definition → Block diagram → Working → Derivation → Applications

For every PLL question, draw the block diagram first. Keep labels far away from wires. Use only straight and perpendicular feedback lines.

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1. Phase 2 Target: What You Must Be Able To Answer

After completing this phase, you must be able to answer these questions by yourself:

1. What is a PLL? Draw and explain its blocks.
2. Explain the roles of phase detector, low-pass filter and VCO.
3. Develop the linear model of a PLL.
4. Derive the closed-loop transfer function of a PLL.
5. Explain free-running, capture and lock modes.
6. Define capture range and lock range.
7. Explain PLL as a frequency demodulator.
8. Explain PLL as a frequency synthesizer using a divide-by- N counter.
9. Solve numerical questions on frequency translation and divide ratio.

Must Know

The PLL is a feedback control system. It tries to make the VCO output frequency equal to the input frequency. The loop is called “locked” when input and VCO output have the same frequency with a fixed phase difference.

2. PLL From Absolute Zero

2.1. Grassroot Idea

A **Phase Locked Loop**, or PLL, is a closed-loop electronic system which compares the phase of an input signal with the phase of a locally generated signal and adjusts the local signal until the two remain locked.

The locally generated signal is produced by a **Voltage Controlled Oscillator**, or VCO.

PLL means phase + frequency tracking system

A PLL has three main blocks:

Phase Detector $\rightarrow$ Low-Pass Filter $\rightarrow$ VCO

The output of the VCO is fed back to the phase detector. Therefore, it is a feedback loop.

2.2. Simple Physical Meaning

Suppose the input frequency increases slightly. Then the phase difference between input and VCO output starts changing. The phase detector senses this error. The low-pass filter converts it into a smooth control voltage. This control voltage forces the VCO frequency to increase. Finally, the VCO catches the input frequency again.

Thus,

PLL continuously corrects the VCO frequency according to phase error.

3. Basic PLL Block Diagram

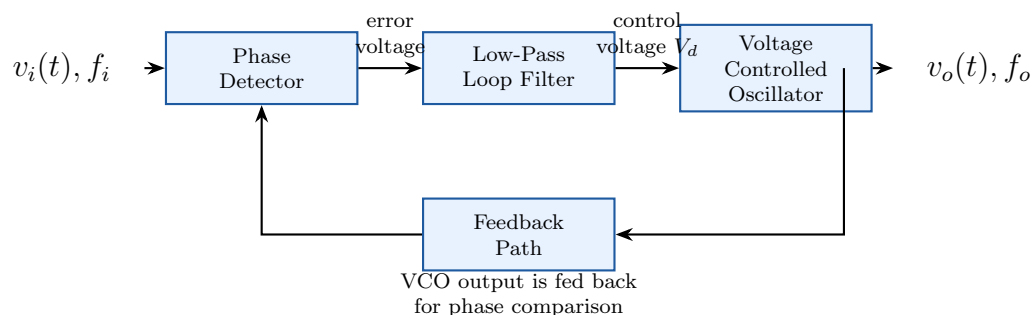


Figure 1: Basic block diagram of a Phase Locked Loop.

3.1. Block-by-Block Explanation

1. Phase Detector:

It compares the phase of the input signal and the phase of the VCO output signal. Its output is an error voltage proportional to phase difference.

$$V_e(t) \propto \phi_e(t)$$

2. Low-Pass Filter:

The phase detector output contains unwanted high-frequency components. The low-pass filter removes these components and gives a smooth DC-like control voltage.

$$V_d(t) = \text{filtered error voltage}$$

3. VCO:

The VCO generates an output frequency controlled by the input control voltage.

$$\omega_o = \omega_{FR} + K_v V_d$$

4. Feedback Path:

The VCO output is fed back to the phase detector. This feedback makes the system self-correcting.

4. Phase Detector: Zero-to-Exam Understanding

4.1. Definition

A phase detector is a circuit whose output voltage depends on the phase difference between two input signals.

Let the input signal phase be

$$\theta_i(t)$$

and the VCO output phase be

$$\theta_o(t)$$

Then the phase error is

$$\phi_e(t) = \theta_i(t) - \theta_o(t)$$

For small phase error, the phase detector is assumed linear:

$$V_e(t) = K_d \phi_e(t)$$

where,

$$K_d = \text{phase detector gain in volt/radian}$$

4.2. Multiplier Type Phase Detector

Let the two sinusoidal signals be

$$v_i(t) = A_i \sin(\omega_i t + \theta_i)$$

and

$$v_o(t) = A_o \sin(\omega_o t + \theta_o)$$

A multiplier phase detector multiplies them:

$$v_m(t) = A_i A_o \sin(\omega_i t + \theta_i) \sin(\omega_o t + \theta_o)$$

Using,

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

we get,

$$v_m(t) = \frac{A_i A_o}{2} [\cos\{(\omega_i - \omega_o)t + (\theta_i - \theta_o)\} - \cos\{(\omega_i + \omega_o)t + (\theta_i + \theta_o)\}]$$

When PLL is near lock,

$$\omega_i \approx \omega_o$$

So the difference-frequency part becomes related to phase error:

$$\frac{A_i A_o}{2} \cos(\theta_i - \theta_o)$$

The sum-frequency part is removed by the low-pass filter.

Phase detector gives phase-error information.

Do Not Write This Mistake

Do not write that the phase detector directly changes frequency. It does not. It only produces an error voltage. The VCO changes frequency.

5. Low-Pass Filter in PLL

5.1. Why LPF is Required

The phase detector output contains:

useful low-frequency error component

and

unwanted high-frequency ripple component

The low-pass filter removes the high-frequency ripple and gives a smooth control voltage.

$$V_d(s) = F(s)V_e(s)$$

where $F(s)$ is the transfer function of the loop filter.

5.2. Role of LPF in Semester Words

1. It removes high-frequency components from the phase detector output.
2. It provides a smooth control voltage for the VCO.
3. It affects the capture range of the PLL.
4. It affects lock stability and transient response.
5. It prevents the VCO from responding to unwanted ripple.

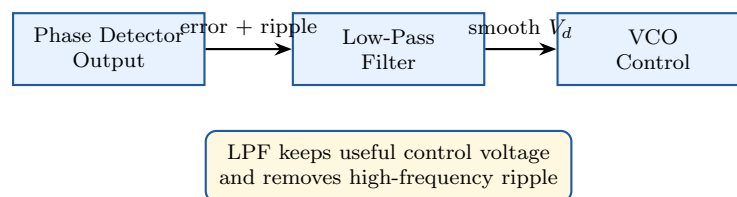


Figure 2: Role of low-pass filter in PLL.

6. VCO: Voltage Controlled Oscillator

6.1. Definition

A VCO is an oscillator whose output frequency depends on the input control voltage.

When no input control voltage is applied, the VCO runs at its free-running angular frequency:

$$\omega_{FR}$$

When control voltage $V_d(t)$ is applied, the output angular frequency becomes:

$$\omega_o(t) = \omega_{FR} + K_v V_d(t)$$

where,

$$K_v = \text{VCO sensitivity in rad/sec/volt}$$

6.2. Phase Relation of VCO

Frequency is the time derivative of phase:

$$\omega_o(t) = \frac{d\theta_o(t)}{dt}$$

Considering deviation from free-running frequency,

$$\Delta\omega_o(t) = K_v V_d(t)$$

But,

$$\Delta\omega_o(t) = \frac{d\theta_o(t)}{dt}$$

Taking Laplace transform,

$$s\theta_o(s) = K_v V_d(s)$$

Therefore,

$$\frac{\theta_o(s)}{V_d(s)} = \frac{K_v}{s}$$

So, in the linear PLL model, the VCO behaves like an integrator.

$$\text{VCO block} = \frac{K_v}{s}$$

7. Linear Model of PLL

7.1. Problem Type

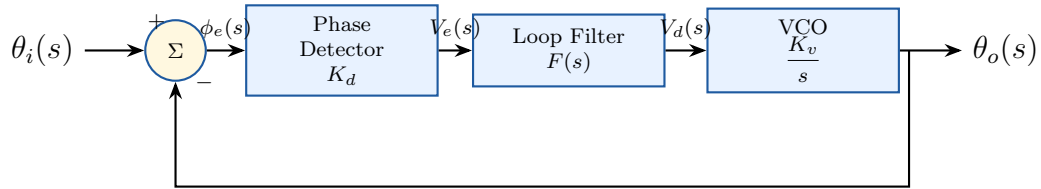
Semester question may ask:

Develop the linear model of a PLL and deduce its transfer function.

Write in this order:

phase error $\rightarrow$ phase detector $\rightarrow$ LPF $\rightarrow$ VCO $\rightarrow$ closed-loop transfer function

7.2. Linear Model Diagram



Negative feedback forces VCO phase to follow input phase.

Figure 3: Linear model of a phase locked loop.

7.3. Step-by-Step Derivation

Phase error:

$$\phi_e(s) = \theta_i(s) - \theta_o(s)$$

Phase detector output:

$$V_e(s) = K_d \phi_e(s)$$

$$V_e(s) = K_d [\theta_i(s) - \theta_o(s)]$$

Loop filter output:

$$V_d(s) = F(s) V_e(s)$$

$$V_d(s) = K_d F(s) [\theta_i(s) - \theta_o(s)]$$

VCO output phase:

$$\theta_o(s) = \frac{K_v}{s} V_d(s)$$

Substitute $V_d(s)$:

$$\theta_o(s) = \frac{K_v}{s} K_d F(s) [\theta_i(s) - \theta_o(s)]$$

Let,

$$\boxed{K = K_d K_v}$$

Then,

$$\theta_o(s) = \frac{KF(s)}{s} [\theta_i(s) - \theta_o(s)]$$

Multiply both sides by s :

$$s\theta_o(s) = KF(s)\theta_i(s) - KF(s)\theta_o(s)$$

Bring output terms together:

$$s\theta_o(s) + KF(s)\theta_o(s) = KF(s)\theta_i(s)$$

Factor out $\theta_o(s)$:

$$\theta_o(s)[s + KF(s)] = KF(s)\theta_i(s)$$

Therefore,

$$\boxed{\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}}$$

This is the closed-loop transfer function of PLL.

7.4. Error Transfer Function

$$\phi_e(s) = \theta_i(s) - \theta_o(s)$$

Divide by $\theta_i(s)$:

$$\frac{\phi_e(s)}{\theta_i(s)} = 1 - \frac{\theta_o(s)}{\theta_i(s)}$$

$$\frac{\phi_e(s)}{\theta_i(s)} = 1 - \frac{KF(s)}{s + KF(s)}$$

$$\frac{\phi_e(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}$$

Must Know

For semester answer, always write both:

$$\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}$$

and

$$\frac{\phi_e(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}$$

8. Modes of Operation of PLL

PLL has three important modes:

Free-running mode Capture mode Lock mode

8.1. Free-Running Mode

When no input signal is applied to the PLL, the phase detector output is zero. Therefore, the low-pass filter output is also zero. The VCO runs at its natural frequency.

$$\omega_o = \omega_{FR}$$

This is called the free-running mode.

8.2. Capture Mode

When an input signal is applied, the VCO frequency starts changing due to the control voltage from the loop filter. The PLL tries to make the VCO frequency equal to the input frequency.

This process of acquiring lock is called capture.

Capture mode means PLL is trying to lock.

8.3. Lock Mode

When the VCO frequency becomes equal to the input frequency, the PLL is locked.

$$f_o = f_i$$

In lock condition, the phase difference becomes constant.

$$\phi_e = \text{constant}$$

8.4. Mode Diagram

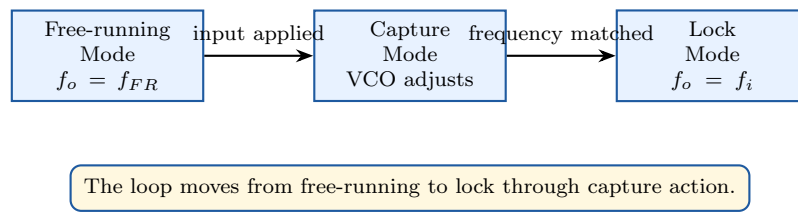


Figure 4: Operating modes of PLL.

9. Capture Range and Lock Range

9.1. Capture Range

The **capture range** is the range of input frequencies over which the PLL can acquire lock starting from the unlocked condition.

$$\text{Capture range} = \text{frequency range for acquiring lock}$$

It depends strongly on the low-pass filter characteristics.

9.2. Lock Range

The **lock range** is the range of input frequencies over which the PLL remains locked after lock has already been achieved.

$$\text{Lock range} = \text{frequency range for maintaining lock}$$

Generally,

$$\text{Lock range} > \text{Capture range}$$

9.3. Diagram of Capture and Lock Range

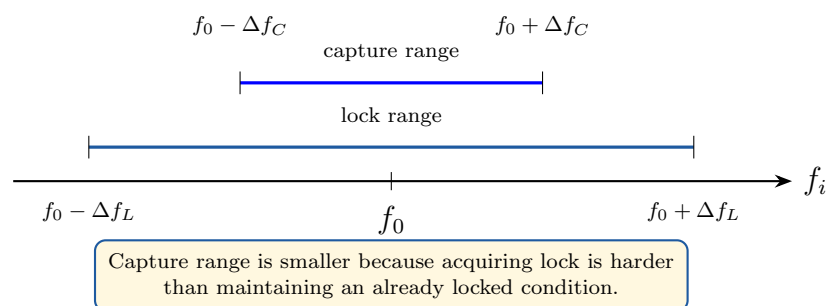


Figure 5: Capture range and lock range of PLL.

Do Not Write This Mistake

Do not confuse capture range and lock range. Capture range is for starting from unlocked condition. Lock range is for remaining locked after lock is already established.

10. PLL as Frequency Demodulator

10.1. Question Pattern

Semester question may ask:

How can a PLL be used as a frequency demodulator?

Write this answer in the following sequence:

FM input $\rightarrow$ PLL locks $\rightarrow$ VCO follows instantaneous frequency $\rightarrow$ control voltage becomes demodulated

10.2. Basic Concept

In an FM signal, the message signal is present in the instantaneous frequency deviation.

Let the FM input angular frequency be

$$\omega_i(t) = \omega_c + k_f m(t)$$

where,

$$m(t) = \text{message signal}$$

$$k_f = \text{FM sensitivity}$$

When PLL is locked, the VCO frequency follows the input instantaneous frequency:

$$\omega_o(t) = \omega_i(t)$$

But VCO frequency is

$$\omega_o(t) = \omega_{FR} + K_v V_d(t)$$

For FM demodulation, the VCO free-running frequency is adjusted to the carrier frequency:

$$\omega_{FR} = \omega_c$$

Therefore,

$$\omega_o(t) = \omega_c + K_v V_d(t)$$

In locked condition,

$$\omega_c + K_v V_d(t) = \omega_c + k_f m(t)$$

Cancel ω_c :

$$K_v V_d(t) = k_f m(t)$$

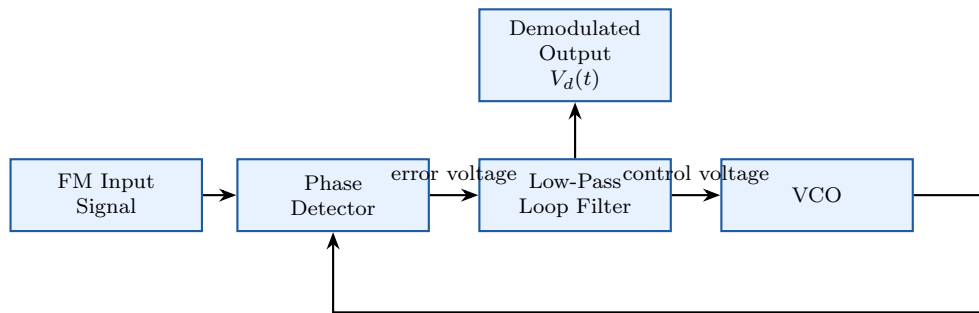
Hence,

$$V_d(t) = \frac{k_f}{K_v} m(t)$$

Therefore, the loop filter output is proportional to the message signal.

$$V_{\text{demod}}(t) = V_d(t)$$

10.3. Diagram of PLL Frequency Demodulator



In lock, VCO follows input FM frequency.
The control voltage is proportional to the original message.

Figure 6: PLL used as frequency demodulator.

10.4. Final Semester Answer

A PLL can be used as a frequency demodulator because in the locked condition the VCO follows the instantaneous frequency of the incoming FM signal. The voltage required to control the VCO is proportional to the instantaneous frequency deviation. Since frequency deviation is proportional to the modulating signal, the loop filter output gives the demodulated output.

$$V_{\text{demod}}(t) = \frac{k_f}{K_v} m(t)$$

11. PLL as Frequency Synthesizer

11.1. Basic Idea

A PLL can generate stable output frequencies using a reference frequency and a frequency divider.

In a frequency synthesizer, the VCO output is divided by N and then compared with the reference frequency.

In lock condition,

$$\frac{f_o}{N} = f_{\text{ref}}$$

Therefore,

$$f_o = N f_{\text{ref}}$$

Thus, by changing N , different output frequencies can be generated.

11.2. Block Diagram

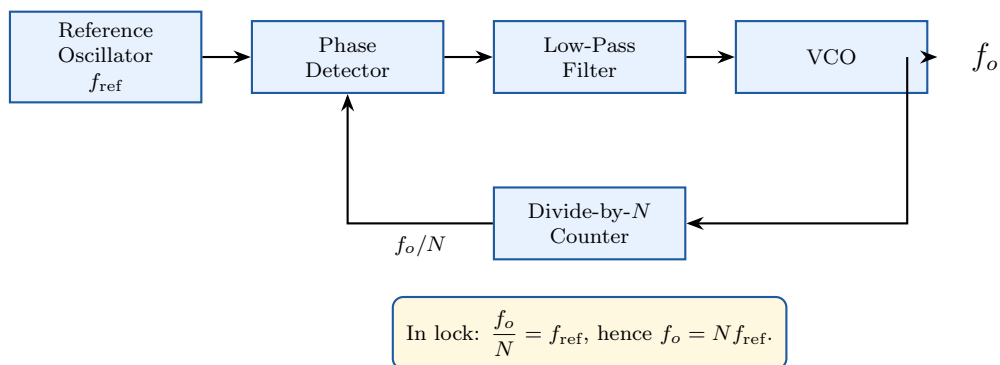


Figure 7: PLL frequency synthesizer using divide-by- N feedback.

11.3. Numerical Pattern

If,

$$f_{\text{ref}} = 50 \text{ kHz}$$

and the counter divide values are

$$N = 2, 4, 8, 16$$

then,

$$f_o = N f_{\text{ref}}$$

N	$f_o = N f_{\text{ref}}$
2	100 kHz
4	200 kHz
8	400 kHz
16	800 kHz

So, if the switch repeatedly makes contacts in the order 1 – 2 – 3 – 4, the synthesizer output frequencies are:

100 kHz, 200 kHz, 400 kHz, 800 kHz

Must Know

The free-running frequency of the VCO is only the initial unlocked frequency. In lock condition, the PLL output frequency is decided by:

$f_o = N f_{\text{ref}}$

12. One-Page PLL Formula Bank

$$\phi_e(t) = \theta_i(t) - \theta_o(t)$$

$$V_e(t) = K_d \phi_e(t)$$

$$V_d(s) = F(s)V_e(s)$$

$$\omega_o(t) = \omega_{FR} + K_v V_d(t)$$

$$\frac{\theta_o(s)}{V_d(s)} = \frac{K_v}{s}$$

$$K = K_d K_v$$

$$\frac{\theta_o(s)}{\theta_i(s)} = \frac{KF(s)}{s + KF(s)}$$

$$\frac{\phi_e(s)}{\theta_i(s)} = \frac{s}{s + KF(s)}$$

$$V_{\text{demod}}(t) = \frac{k_f}{K_v} m(t)$$

$$f_o = N f_{\text{ref}}$$

13. How to Recognise PLL Questions in Semester

Question Language	What You Must Write
“List stages of PLL”	Draw block diagram and explain phase detector, LPF, VCO.
“Linear model of PLL”	Draw $\theta_i(s)$, K_d , $F(s)$, K_v/s , feedback, then derive transfer function.
“Capture range and lock range”	Define both, draw frequency-axis diagram, write lock range is usually larger.
“PLL as frequency demodulator”	Write FM input frequency, VCO frequency, locked condition, derive $V_d(t) = \frac{k_f}{K_v}m(t)$.
“PLL synthesizer”	Draw divide-by- N feedback diagram and use $f_o = Nf_{\text{ref}}$.
“Role of LPF”	Write: removes sum-frequency/ripple, gives smooth control voltage, affects capture and stability.

14. Exam Writing Templates

14.1. Template 1: Explain PLL Blocks

Semester Command

Start with definition, then draw block diagram, then explain three blocks. End with lock condition.

Answer skeleton:

A Phase Locked Loop is a closed-loop feedback system which locks the phase and frequency of a VCO output signal with the phase and frequency of an input signal. It consists of a phase detector, low-pass filter, VCO and feedback path.

The phase detector compares input phase and VCO phase and gives error voltage. The low-pass filter removes high-frequency components and gives smooth control voltage. The VCO changes its frequency according to this control voltage. In lock condition,

$$f_o = f_i$$

and the phase difference remains constant.

14.2. Template 2: Linear Model

Semester Command

Never skip the VCO integrator step. Examiner expects:

$$\omega_o = \frac{d\theta_o}{dt} \Rightarrow \frac{\theta_o(s)}{V_d(s)} = \frac{K_v}{s}$$

14.3. Template 3: Frequency Demodulator

Semester Command

Always show:

$$\omega_i(t) = \omega_c + k_f m(t)$$

$$\omega_o(t) = \omega_c + K_v V_d(t)$$

Then equate in lock condition.

15. Final Rapid Revision

15.1. If only 20 minutes are left

Revise only these:

1. Basic PLL block diagram.
2. Linear model transfer function.
3. Capture range vs lock range.
4. PLL frequency demodulator derivation.
5. Frequency synthesizer formula $f_o = Nf_{\text{ref}}$.

15.2. Most Probable Diagram Commands

1. Draw PLL block diagram with feedback path below.
2. Draw linear model with $\theta_i(s)$, summing junction, K_d , $F(s)$, K_v/s .
3. Draw PLL demodulator with demodulated output taken from LPF.
4. Draw PLL synthesizer with divide-by- N counter in feedback.
5. Draw capture and lock range on frequency axis.

Do Not Write This Mistake

Do not write long paragraphs without a diagram in PLL answers. In semester paper, PLL answers must be block-diagram based. Draw first, explain next, derive last.

Phase 2 Completed: PLL Mastery